

Data Engineering and AI (ENGE707)



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Rationale

Why did you enrol in the Data Engineering and AI course?



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Rationale

Why did you enrol in the Data Engineering and AI course?



- To learn how *data engineering* supports *AI systems*.
- To build skills in *data handling, cleaning, visualisation*, and *machine learning*.
- To understand the *AI pipeline* from data to model deployment.
- To explore *modern AI approaches* like deep learning and transformer models.
- To learn about *ethical and cultural issues* in AI, including bias and *Māori data sovereignty* in Aotearoa New Zealand

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Rationale

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG PILE OF LINEAR ALGEBRA, THEN COLLECT THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL THEY START LOOKING RIGHT.



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Teaching Team

Abubakar Siddique

Dr. Siddique develops cognitive-neuroscience-inspired machine learning systems for areas such as computer vision, navigation, and bioinformatics. He has received several awards, including the Emerging Research Excellence Medal and VUWSA Gold Award. Before academia, he was a Principal Software Engineer, leading enterprise projects for clients such as Xerox, IBM, and Adobe.



Anubha Kalra

Anubha Kalra is a Senior Lecturer in the Electrical and Electronics Department at Auckland University of Technology (AUT). Before this, she was a research fellow in biomedical engineering at the Institute of Biomedical Technologies, AUT, in New Zealand. Her research spans the fields of Electrical Engineering, Biomedical Engineering, and Rehabilitation.



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Teaching Team

Abdul Rehman Ahmed



Ayesha Ajmal



Minh Quach



Adebanji Adeleye



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Learning Outcomes

On successful completion of this course, students will be able to meet these *learning outcomes* (LOs):

- **LO1:** Elicit requirements for a data engineering project
- **LO2:** Handle and visualise data appropriately for an application
- **LO3:** Select and describe machine learning/AI algorithms appropriate for an application
- **LO4:** Apply a machine learning algorithm from a toolbox/API
- **LO5:** Judge the validity of the results

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Course Delivery

Delivery	Day	Time	Room
Lecture	Friday	10:00 – 12:00	WS114
Lab (W201)	Thursday	12:00 – 14:00	WS313-314
Lab (W202)	Monday	08:00 – 10:00	WS220

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Assessments

Assessment	Weighting	Learning Outcomes
Mid-Project Report (<i>Scoping Document</i>)	30%	1,2,3
Final-Project Report, Demo, and Presentation	30%	1,2,3,4,5
ML Assignment (<i>Kaggle Competition</i>)	40%	1,2,3,4,5

Surprise Quizzes

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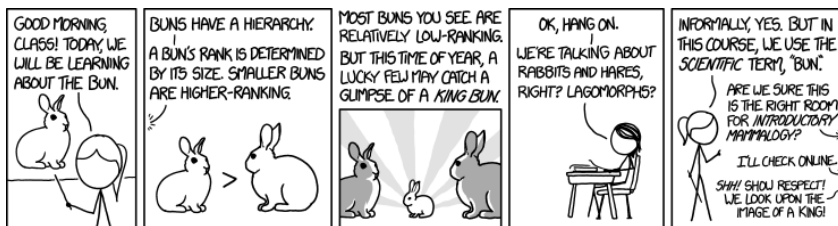
Assessments

- You are encouraged to use *AI-assisted tools* such as GitHub Copilot and LLMs for *coding* and learning.
- However, you must *fully understand* your own work and be able to explain it.
- Commit and push* your work regularly to the *Git repository* to show your progress.
- All assessment submissions must be *your own original work*.
- Descriptive answers generated by *LLMs (AI)* are *strictly prohibited*.
- Assessments must be submitted by the dates and times announced by the lecturer.

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Teaching Strategy



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Teaching Strategy

- The course will be delivered through a combination of *lectures*, *labs*, and *self-study* material.
- Lab activities are designed to build *deep understanding* and provide hands-on experience.
- For complex concepts, *toy examples* and *step-by-step guides* will be provided.
- Assignments will help you apply your knowledge to real-world problems.
- Lecturers and TAs will be available during labs to *demonstrate techniques*, explain concepts, and *provide guidance*.
- Learning sessions will be *interactive*, and you are *encouraged to ask questions* in lectures, labs, and Q&A forums.

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Threshold Concepts



A **threshold concept** can be considered as akin to a portal, opening up a new and previously inaccessible way of thinking about something. It represents a transformed way of understanding, or interpreting, or viewing something without which the learner cannot progress.

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Text & Tools

There are no required textbooks for this course

Recommended books

- Data Science from Scratch by Joel Grus
- Python for Data Analysis by Wes McKinney
- Fundamentals of Data Engineering by Joe Reis & Matt Housley
- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron

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Text & Tools

Recommended Tools

- Python
- Scikit-learn
- Git and GitHub
- NumPy, Pandas
- Matplotlib and Seaborn
- TensorFlow or PyTorch
- Google Colab /Anaconda/Jupyter Notebook / JupyterLab
- AI-assisted coding tools such as GitHub Copilot or ChatGPT.

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“Where shall I begin?” The White Rabbit asked.
 “Begin at the beginning” the King said,
 “and go on till you come to the end: then stop.”



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**Thank You
&
Questions**



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